Tamajeet Biswas

Bachelor of Technology
in Electrical Engineering
Indian Institute Of Technology, Ropar

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Education

Degree	Institute/Board	CGPA/Percentage	Year
Bachelor of Technology	Indian Institute of Technology, Ropar	7.11 (Till 6th Sem)	2023-PRESENT
Senior Secondary	Central Board of Secondary Education	91%	2022
Secondary	Central Board of Secondary Education	95.6%	2020

Technical Skills

- **Programming & Scripting Languages**: Verilog (HDL), VHDL, Python, C, C++, Arduino IDE
- **Hardware & Microcontrollers**: Arduino UNO (ATmega328P), ESP32, ESP8266 (Wi-Fi SoCs), Mimas A7 Mini FPGA Board
- **Tools, Frameworks & Libraries**: Visual Studio Code (VS Code), Xilinx Vivado, ModelSim, GTKWave, Icarus Verilog, Firebase
- **Design & Simulation Tools**: Blender, TinkerCAD, Canva

Experience

- **Apsis Solutions** *June 2025*
  VLSI Intern **Remote**
  - Executed 3 RTL design projects in Verilog/VHDL using Xilinx Vivado & ModelSim for FPGA prototyping and simulation.
  - **Traffic Light Control System (Mimas A7 Mini)** **Github : github.com/Tamajeet7/Traffic-light-control**
    Real-time traffic light controller using FSMs in VHDL; validated on FPGA hardware (Mimas A7 Mini).
  - **FSM Sequence Detector (Verilog)** **Github : github.com/Tamajeet7/Sequence-Detector**
    Designed Mealy and Moore FSMs for sequence detection; verified accuracy with ModelSim.
  - **16-bit RISC Processor (Verilog)** **Github : github.com/Tamajeet7/16-Bit-RISC-Processor**
    Implemented 16-bit processor with ALU, register file, and control unit; validated with modular testbenches.

Projects

- **MIPS32 Processor Using Pipelining** *Jan 2025*
  Verilog Course under Dr. Indranil Sengupta **GitHub : github.com/Tamajeet7/MIPS32-Processor-Pipeline**
  - Implemented a high-performance MIPS32 processor in Verilog using a 5-stage pipelined architecture with instruction fetch, decode, execute, memory access, and write-back stages. Supported arithmetic, branching, load/store, and halt operations.
  - Performed extensive verification using custom test cases to ensure instruction accuracy, hazard detection/forwarding, and pipeline efficiency. Validated design through simulation and waveform analysis.
- **Smart Parking System (IoT + Firebase)** *April-May 2025*
  Team Project | Role: ESP & Firebase Developer **GitHub : github.com/Tamajeet7/Smart-Parking-System**
  - Developed an IoT-based smart parking system using ESP8266 microcontrollers for real-time slot detection, gate automation, and hardware control.
  - Integrated IR sensors, servo motor, and additional ESP modules to enable LED/buzzer alerts for occupancy status.
  - Implemented Firebase Realtime Database for cloud-based slot management, booking updates, and automated gate control logic, ensuring reliable data synchronization between hardware and mobile interface.

Relevant Coursework

- **Electrical Engineering**: Digital Circuits, Digital Logic Design, Basic Electronics,Embedded Systems, Signals and Systems, Circuit Theory, Control Systems, Materials Science for Electrical Engineers, Electromechanics
- **Specialized Courses**: Hardware Modelling Using Verilog (Dr. Indranil Sengupta)
- **Computer Science and Engineering**: Introduction to Computing and Data Structures, Tinkering Lab
- **Mathematics**: Probability and Stochastic Processes, Advanced Calculus, Linear Algebra, Differential Equations

Positions of Responsibility

- **Class Lead**, Pehchaan Ek Safar, IIT Ropar *Sept. 2024 – Apr. 2025*
- **Social Media Team Member**, Pehchaan Ek Safar, IIT Ropar *Sep. 2023 – Present*

Honors & Awards

- **Reliance Foundation Undergraduate Scholar**, National-level aptitude test *2024*